Investigating the Impact of Training Protocols on Myoelectric Pattern Recognition Control in Upper-Limb Amputees

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***Abstract*— Myoelectric control schemes, pivotal in the control of prosthetic limbs, are often developed and evaluated in ideal laboratory conditions. However, these controlled environments may not fully represent the diverse challenges users face in real-world scenarios. The present study aims to tackle some of the existing research limitations by exploring the influence of various model training protocols on myoelectric pattern recognition within a semi-autonomous control system, which has been shown to reduce user cognitive load and enhance overall system performance. Specifically, we focus on the effects of limb movement and weight-bearing activities. We investigate the effect of four distinct training protocols in pattern recognition control for upper-limb prostheses, including training without a prosthetic hand, training with a prosthetic hand and static gestures, training with a prosthetic hand and dynamic movements guided by a graphical user interface (GUI), and training with a prosthetic hand having dynamic transfers and unguided. By examining these conditions, we aim to provide an understanding of how different training protocols and different labeling methods influence myoelectric pattern recognition control. Our results, based on experiments conducted with 14 non-disabled and one amputee participant, suggest that introducing the weight of the prosthetic hand and dynamic movements of the arm to the training data improves the accuracy and robustness of the control scheme. Real-time control experiments with a group of five non-disabled and one amputee participant using a multi-DOF prosthetic hand also verify our findings.**

***Index Terms*— Myoelectric control, pattern recognition, sEMG, training protocol, transradial prosthetic.**

I. Introduction

PROSTHESES are vital to enhancing the quality of life for a large population of approximately 65 million amputees worldwide as of 2020, with an estimated 1.5 million amputations occurring annually [1]. There has traditionally been a high rejection rate and low satisfaction with prostheses among amputees, especially upper-limb amputees [2]. Predicting user intent from biological signals, particularly Electromyography (EMG), the most important means of controlling upper-limb prostheses, has been a tough challenge. Despite decades of advancements in myoelectric prostheses control, challenges persist, prompting ongoing research to improve their effectiveness. One notable approach is myoelectric pattern recognition (PR), which relies on separable and distinct muscle contraction patterns. While PR controllers have shown promise in controlled laboratory settings, their translation to clinical practice faces hurdles [3]. A critical issue lies in ensuring the generalization ability of learned classifiers across diverse real-world application environments. Factors such as electrode shifting [4], varying contraction amplitude levels, and changing body positions [5] introduce signal variations, impacting the performance of classifiers trained in ideal laboratory conditions [3], [6]. While contemporary research has predominantly focused on improving classifier accuracy and feature optimization using controlled training datasets, and mostly in offline scenarios, several studies have investigated one or more of these confounding factors, exploring solutions like dynamic and multi-position training, sensor fusion, and advanced classification algorithms to mitigate their effect. Limb position changes is one of the factors that has been extensively studied. For instance, Fougner et al. [7] reduced classification errors by integrating accelerometer-based position detection with multi-position training, while Liu et al. [8] introduced a multi-condition training scheme and a Conditional Gaussian Mixture Model to mitigate errors from dynamic arm movements. Beaulieu et al. [9] demonstrated that training across multiple limb positions significantly reduces classification errors, while Geng et al. [10] found that multiposition classifiers improved motion completion rates compared to single-position classifiers in real-time testing. Gigli et al. [11] investigated a dynamic training protocol that utilized EMG signals captured during continuous limb movements encompassing all target motion patterns. Their findings highlighted the benefits of the dynamic training protocol, including improved controllability and a less exhausting data collection process. More recent approaches, such as recurrent convolutional neural networks [12] and

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Interdisciplinary Committee on Ethics in Human Research (ICEHR), under Application No. 20210316-SC, and performed in line with the Declaration of Helsinki.

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weighted recursive Gaussian processes [13], have further improved robustness to positional variations. While limb position has been extensively studied, other factors such as electrode shifts, varying contraction levels, and external loads have received comparatively less attention. Scheme and Englehart [14] investigated the impact of contraction strength on the robustness of pattern recognition-based myoelectric systems. They employed a ramp training scheme where contraction intensity increases dynamically from a light to a hard level, and demonstrated that this training approach outperforms all level-based schemes. Wu et al. [15] addressed electrode shifts by proposing a shift-robust convolutional neural network, while Ameri et al. [16] introduced an innovative self-recalibrating system leveraging transfer learning with convolutional neural networks to mitigate the impact of electrode shifts. Wang et al. [17] investigated the combined effects of limb positions and electrode shifts, demonstrating that these factors create statistically distinct class instances in the feature space, leading to significant classification performance degradation. The study by Cipriani et al. [18] was one of the earliest studies to explore the impact of prosthetic weight and dynamic movements on pattern recognition-based control systems, demonstrating that these factors significantly degrade classification performance, particularly for the relax state. However, their work was limited to offline testing with non-disabled participants, leaving the effects on real-time control and amputee populations unexplored. Yang et al. [19] developed dynamic training protocols incorporating force variations and limb positions, showing improved generalization in dynamic scenarios.

Teh and Hargrove [20] studied the effect of limb position and external load in an offline setting and showed that both factors independently degraded classification performance in non-disabled subjects, while interaction effects were observed in amputees. Multi-position and dynamic training reduced the effects of limb position and improved performance but did not eliminate load effects. In another study [21], the authors examined the same factors in a real-time setting using a VR-based Target Achievement Control test. Results showed that limb position significantly impacted performance, while external load effects were less pronounced. Dynamic training reduced inter-position variability in non-disabled participants and partially mitigated limb position effects in amputees; however, it did not improve performance under external load conditions.

While progress has been made in addressing specific confounding factors, many focus on single variables or offline evaluations, which was previously shown to have minimal correlation with real-time usability [22], leaving the combined impact of multiple real-world factors in prosthetic control insufficiently explored. Moreover, limited attention has been given to understanding how training protocols and data labeling strategies affect robustness and usability in scenarios involving both prosthetic weight and dynamic movements, which are unavoidable in practical applications.

To bridge this gap, this study investigates how different training protocols influence the robustness and generalization of myoelectric control in a semi-autonomous prosthetic system, where grasp types are determined using a wrist-mounted camera, and EMG signals detect the user's intent to execute the movement. We focus on the effects of prosthetic hand's weight, arm movement, guided and unguided data collection, and manual vs. automatic labeling on classifier performance and real-time usability. To systematically evaluate these factors, we designed four training protocols: without the prosthetic hand and without arm movement, wearing the prosthetic hand and without arm movement, and two sessions of collection wearing the prosthetic hand and with arm movement, one guided through a GUI and the other one unguided. Using intra-protocol and inter-protocol validation, we assess the repeatability and generalization ability of each training protocol, followed by an evaluation of their real-time performance. The insights from this study contribute to a broader understanding of how training methodologies affect myoelectric control robustness and usability in practical applications and if offline performance is a reliable predictor of real-world prosthetic performance. The overall experiment pipeline is shown in Fig. 1.

II. Materials and Methods

A. Subjects

Fourteen non-disabled subjects (eleven males and three females, right-handed, ages 24-37) and one bilateral transradial amputee (male, left-handed, 55-year-old, amputation caused by electric shock) participated in our experiment. The length of the amputee participant's stump is approximately 12 cm. He has been using body-powered prosthetics for almost 17 years since his amputation and has experience using myoelectric prosthetics.

All participants involved in this study provided informed consent before participating. The consent process was conducted in accordance with the ethical standards outlined by the Memorial University of Newfoundland Interdisciplinary Committee on Ethics in Human Research. Participants were informed about the purpose and nature of the study. They were assured of confidentiality and the voluntary nature of their participation. Additionally, participants were informed of their right to withdraw from the study at any time without consequence.

B. Experimental Setting

Surface Electromyographic (sEMG) signals were recorded from participants using eight equidistant electrodes in MINDROVE armband (MindRove Kft., Győr, Hungary). The prosthetic hand employed in these experiments is the Psyonic Ability Hand [23],which provides six degrees of actuation and supports up to 32 distinct grip patterns. It is equipped with a motor-based collision detection mechanism that halts finger movement in the pre-collision direction upon detecting increased motor load, with a default collision threshold set at 3.0V. Additionally, the ability hand is capable of a maximum lifting force of 50 lbs, and individual motions can be completed in as fast as 0.2 seconds. For the experiments in which video recording is used as the ground truth, videos

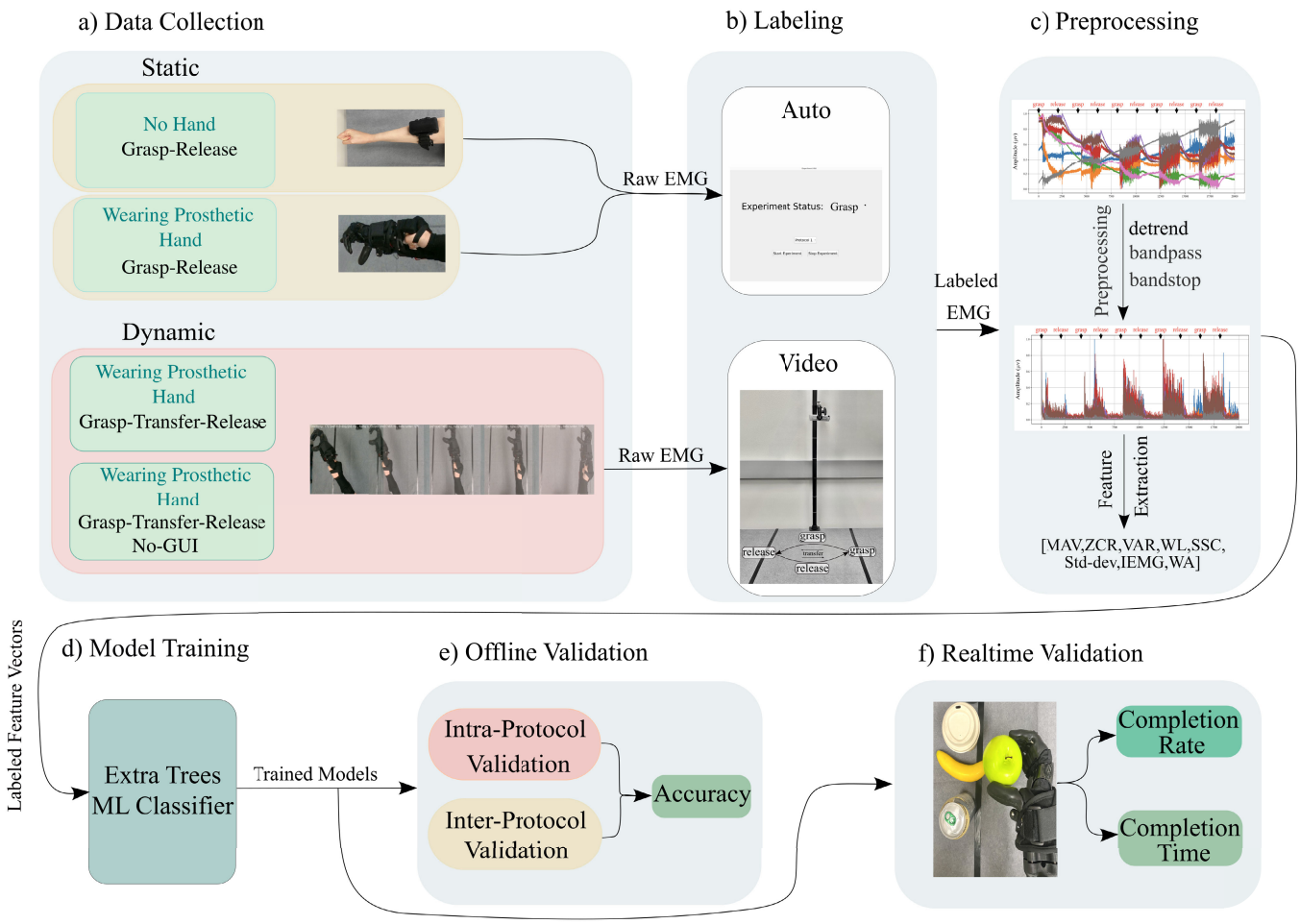


Fig. 1. Overview of the general pipeline employed in this study.

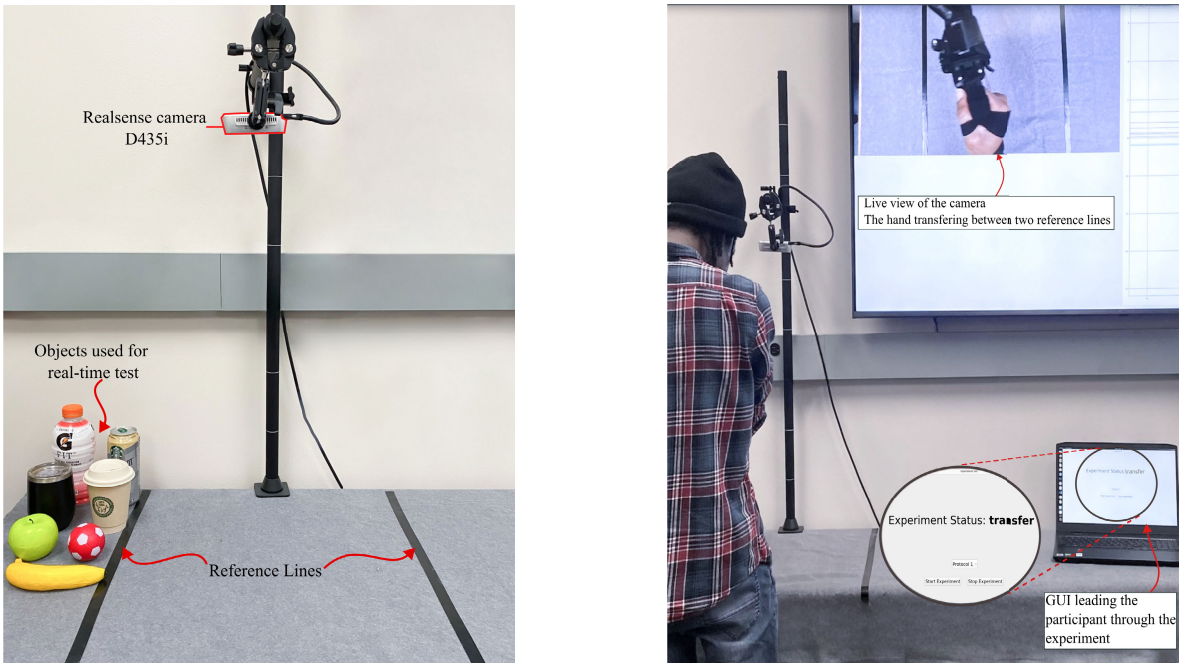
are recorded using a top-view Realsense camera (Fig. 2a). For all the experiments except one, the participants are guided to perform the gestures through a GUI, which shows the gesture to be performed for the duration it should be performed. A beep sound also accompanies the change of gestures to keep the participants more focused. The experimental setup, sensor connectivity, and data collection are managed through the Robot Operating System (ROS). Data from different sensors are collected into ROSbags which guarantees synchronized timestamps. This setup enables precise synchronization across all sensors and the GUI by providing accurate timestamping for each sample.

1) Intact Limb Subjects: For the healthy subjects, the armband was worn around their right forearm. Except for one experiment, participants were wearing the prosthetic hand. A prosthetic simulator was designed to enable healthy volunteers to wear the prosthetic hand as shown in Fig. 3a. This simulator consists of a 3D-printed handle with a forearm support, which is secured to the participant's forearm using a band. The handle-like component is not intended to be grasped by the participant; rather, it serves as a reference point for hand positioning and limits finger movements, ensuring consistency in the experimental gestures. The real-time test is also conducted using the Psyonic hand. The Psyonic hand weighs 490 grams, and including the handle, the whole setup weight is almost 900 grams.

2) Amputee Subject: For the amputee participant, the myo armband was worn around his left stump (Fig. 4a). A prosthetic liner was worn on top of the armband to help keep the armband in place and to add more comfort and cushioning between the stump and the socket (Fig. 4b). The customized socket is then used for wearing the prosthetic hand (Fig. 3b).

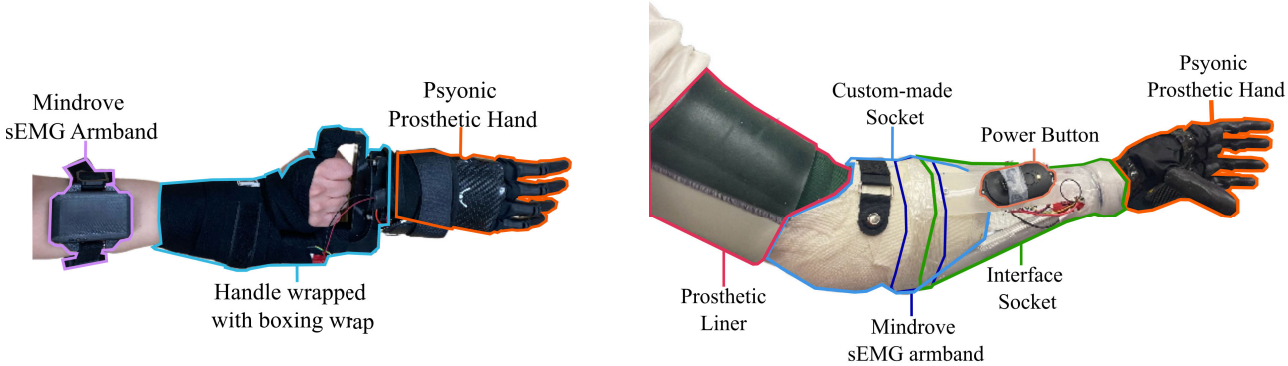
3) Signal Acquisition and Processing: EMG signal is recorded using an 8-channel MINDROVE commercialized armband at 500 Hz frequency. Using this circle-type configuration of the armband makes the setting simple and repeatable and alleviates the need for targeted placement of the electrodes. On the other hand, using this 8-channel armband offers detailed muscle activity patterns, improving generalization across users and conditions. Additionally, since this study examines the impact of prosthetic hand and arm movements, the extra channels help capture subtle EMG signal differences caused by these movements in different muscles. This detailed data is crucial for understanding and mitigating their effects on model performance, ensuring more reliable and stable real-world applications.

The recorded EMG signal was preprocessed (detrend, bandpass, and bandstop filtering) before feature extraction.



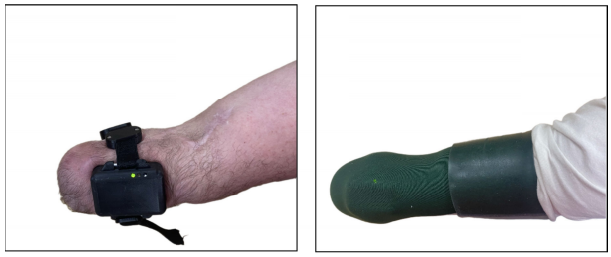
(a) (b)

Fig. 2. **a)** Experiment setup **b)** Healthy Volunteer wearing the Prosthetic hand, doing the experiment. The participant is guided using the GUI to perform grasp, transfer, and release.



(a) (b)

Fig. 3. The setup used for **a)** intact limb subjects and **b)** amputee subject.



(a) (b)

Fig. 4. **a)** EMG armband placement on the stump of amputee subject **b)** Prosthetic liner worn on top of the armband to keep the armband in place and to add a cushioning layer between the residual limb and the socket.

To optimize classification accuracy, we identified an effective subset of time-domain features through an exhaustive feature selection process, ultimately choosing a set of nine features (Mean Absolute Value, Waveform Length, Zero Crossing, Slope Sign Change [24], Variance, IEMG, Wilson Amplitude [25], Standard Deviation and RMS [26]). We selected time-domain features for their computational efficiency and effectiveness in capturing key aspects of muscle activation patterns, making them particularly suitable for real-time EMG classification [27], [28]. The features were extracted from each channel using 400-millisecond (200 samples) windows with a stride of 200 milliseconds (100 samples) (Fig. 1c).

These features were then utilized to train Support Vector Machine (SVM) and Extra Trees classifiers (Fig. 1d). SVM stands out as one of the most prevalent EMG classification techniques, demonstrating exceptional performance [29]. Extra Trees classifier has also been employed in the field of EMG classification recently and exhibited remarkable accuracy [30].

C. Experimental Protocol

1) Training Protocols: Participants completed four sessions of data collection corresponding to four training protocols,

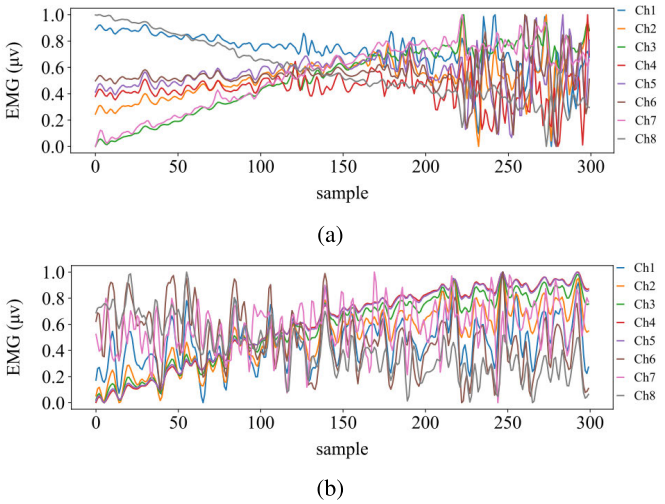


Fig. 5. Exemplary EMG signal from a participant when the hand goes from initial state to grasp **a)** without and with **b)** wearing the prosthetic hand. In (a), the EMG signals exhibit less variability and lower amplitude fluctuations across channels, indicating a more stable muscle activity pattern. Conversely, in (b), the EMG signals show increased variability and higher amplitude fluctuations, reflecting the added complexity and adjustments in muscle activity required to support the prosthetic hand weight.

performing two gestures—grasp and release—to convey their intent in a semi-autonomous control system. In this setup, the system uses a computer vision module to identify the appropriate grasp type based on the object, while the EMG signal captures the participant's intention to either grasp the object or place it. These gestures were performed under four different conditions, each representing a unique training protocol (Fig. 1a). There was a 5-minute rest between each two sessions to minimize the effect of muscle fatigue on the signals. The armband was not taken off from the participant for the whole duration of the experiment. Fig. 2b shows a participant doing the experiment. The training protocols are as follows:

1) **Static No Prosthetic Hand (SNH)**
For the first training protocol, the participants only wore the EMG armband. They were advised to hold the arm at a 90-degree angle and perform grasp and release at the time and for the duration guided by the GUI, as naturally as they could. There were ten repetitions of grasp-release, and each gesture was performed for four seconds, resulting in 80 seconds for the whole session.
2) **Static With Prosthetic Hand (SWH)**
In the second training protocol, participants wore both the EMG armband and the Psyonic hand. The remaining aspects of the experiment mirrored those of the first protocol. The objective of this protocol is to examine the impact of wearing the prosthetic hand and its weight on the EMG signals and, subsequently, on the performance of the models. The exemplary EMG signal from these two protocols is shown in Fig. 5.
3) **Dynamic With Prosthetic Hand - Guided (DWHG)**
For the third training protocol, the participants wore the armband and Psyonic hand. They were asked to stand in front of a table and perform the shown gestures on the GUI for the duration shown. In this experiment, besides grasp and release gestures, transfer was also one of the gestures, where the participant had to move the arm between two reference lines while holding the performed gesture (grasp or release). This experiment is also recorded using a top-view camera to be used as a ground truth to label the signals. This protocol is tailored to investigate the influence of dynamic arm movements on model performance and its generalizability. Additionally, it aims to compare the effectiveness of auto and video labeling strategies.
4) **Dynamic With Prosthetic Hand - Unguided (DWHU)**
This protocol is the same as the previous protocol, except for the GUI. No guidance was provided to the participants. They had to perform each gesture at a reference line and transfer to the other line at their own pace while holding the gesture. The experiment is recorded using a top-view camera and labeled using the video. The objective of this protocol is to assess whether unguided training can positively influence the overall model performance.

2) Protocols for Amputee Subject: For the amputee participant, given our prior knowledge from experiments with healthy participants, we focused on three protocols: SNH as the baseline, SWH, and DWHG (auto-labeled), and omitted the video-labeled protocols. In the SWH protocol, the subject did not wear the prosthetic hand and only used the EMG armband (Fig. 4a). For the rest of the experiments, the subject was equipped with the prosthetic hand. The amputee was instructed to perform muscle contractions, either static or dynamic, as required, while using motor imagery to simulate grasping an object. The rest of the experiments were conducted in the same manner as with non-disabled participants.

3) Automatic Vs. Manual Labeling: To explore the reliability of auto labeling and assess the practicality of incorporating a video ground truth, we examined two labeling methodologies for the DWHG training protocol: automatic labeling, and manual labeling using video analysis. In the auto-labeling protocol, EMG signals are automatically labeled with the corresponding gesture displayed on the GUI, assuming simultaneous execution by the participant. The synchronization between EMG signals and gesture labels is rigorously maintained through timestamp matching, ensuring that each EMG reading aligns precisely with the intended gesture displayed on the GUI. To further enhance the reliability of the automatic labeling, trial runs were conducted to confirm that participants could consistently execute the gestures as instructed. Conversely, in the video-labeling strategy, signals are labeled based on the initial visible change in hand movement at the onset of each gesture (Fig. 1b). This approach aims to mitigate the impact of reaction time on labeling accuracy and assess the reliability of auto-labeling.

4) Model Training and Validation: Our experiments encompass both offline (Fig. 1e) and online (Fig. 1f) analyses. In these analyses, we employ Extra Trees as the classifier. All analyses are subject-dependent, meaning the data used for both training and testing comes from the same subject. During the offline analysis, we validate our trained models through intra and inter-protocol assessments. For intra-protocol validation,

each model undergoes testing on data collected within the same protocol as the training data. Conversely, for inter-protocol validation, each model is assessed using all data collected from the remaining training protocols. To verify our offline findings and compare the real-time performance of different training protocols, we conducted a pick-and-place task using the Psyonic prosthetic hand. This task involved a group of five participants (two female, three male, all right-handed) and the amputee participant. A total of 7 objects (Fig. 2a) with different shapes, textures, and weights had to be transferred from one side of a reference line to the other side of the other one. The distance between the reference lines is 55cm. The original control system designed for the experiment is based on a semi-autonomous control scheme wherein computer vision is responsible for object detection and, consequently, the grasp type determination, and the EMG classifier is responsible for the user intention detection and triggering the grasp or release process. However, as in this study, we only wish to evaluate the performance of EMG classification, we fix the grasp type to "Power Grasp" to avoid any potential delay or error associated with the computer vision module in our analysis. All the objects are chosen so that they can be grasped using a power grasp.

III. Results

From each protocol, we trained two different classifiers (SVM with RBF kernel, and Extra Trees with estimators=200) and validated them on each intra-protocol and inter-protocol, respectively. As the results show no significant difference in classifiers performance, with Extra Trees being slightly better (Fig. 7), we proceed with the Extra Trees results only. In the online analysis, we selected three models trained using the SNH, SWH, and DWHG training data to perform the real-time control of the Psyonic hand to do the pick and place task. As the offline analysis showed that there is no advantage in the video-labeled strategies we skipped those protocols for the real-time test.

A. Auto Labeling Vs. Video Labeling

To compare and validate the performance of the two labeling methodologies, we initially trained a classifier using the data from the DWHG protocol labeled using each methodology and subsequently evaluated the trained models across all training protocols. As Fig. 6 shows, only a marginal improvement in accuracy was observed for the DWHU and SNH protocols when utilizing video labeling, whereas a decline was evident for the SWH protocol. Statistical analysis indicated no significant disparity in classification performance between the two labeling methods. Therefore, in the subsequent analysis, we only consider the auto-labeling protocol.

B. Intra-Protocol Validation

The intra-protocol analysis involved the application of a 10-fold cross-validation approach, with each fold representing a trial of data within a specific training protocol. The resulting average accuracy for each protocol over all participants was computed and is shown in Fig. 7. The variability among

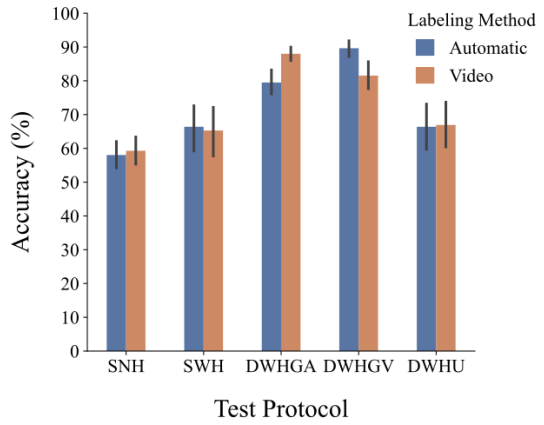


Fig. 6. Comparison of test accuracy between models trained on auto-labeled data and video-labeled data. The X-axis represents the protocols used for collecting the test data. The blue bars (first bar) indicate the accuracy of models trained on auto-labeled data, while the brown bars show the accuracy of models trained on video-labeled data. The automatic labeled accuracy for DWHG (DWHGA) and the video labeled accuracy for DWHG (DWHGV) are calculated using 10-fold cross-validation.

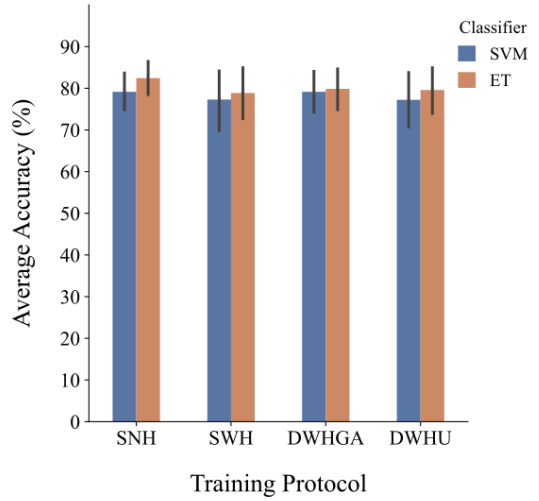


Fig. 7. Intra-Protocol accuracy for all 14 participants recruited for offline analysis using SVM and ET classifiers.

participants can be observed through the boxplot displayed in Fig. 8. These intra-protocol assessments aimed to characterize the inherent strengths and limitations of the classifiers when trained and tested within the same protocol. In intra-protocol validation, the first protocol (SNH), which is also the simplest one, receives the highest average accuracy (83%), which suggests data collected under this protocol may have the strongest repeatability. The protocols involving arm movement and prosthetic hand weight (SWH, DWHG, DWHU) have lower intra-protocol accuracy due to their complex nature.

C. Inter-Protocol Validation

Using only the intra-protocol validation cannot ensure the classifier's generalization ability over the other data-acquisition conditions. So, we validated each protocol over the data collected using the rest of the training protocols, to mimic different real-life scenarios. As can be seen in Fig. 9, the minimum accuracy is for training using the DWHU protocol and testing on the SWH where the accuracy drops from 80%

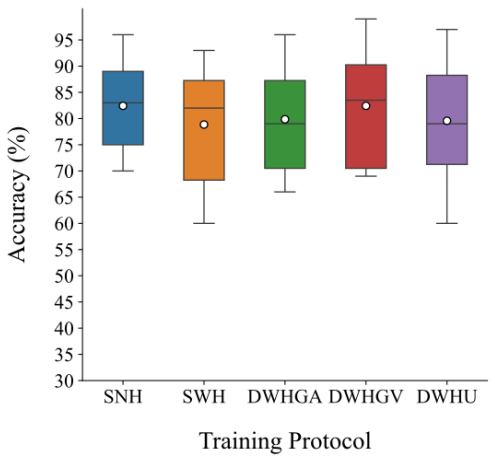


Fig. 8. Intra-Protocol accuracy boxplot over all participants.

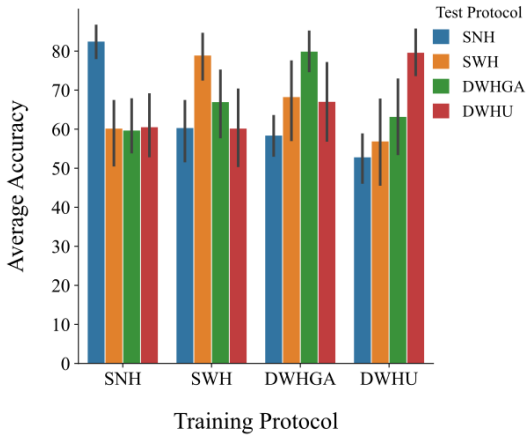


Fig. 9. Validated performance of the classifiers. Bars in the graph are separated into four groups, each indicating an independent training protocol. Bars in different colors (or orders in the group) indicate the accuracy of the related protocol as a test set.

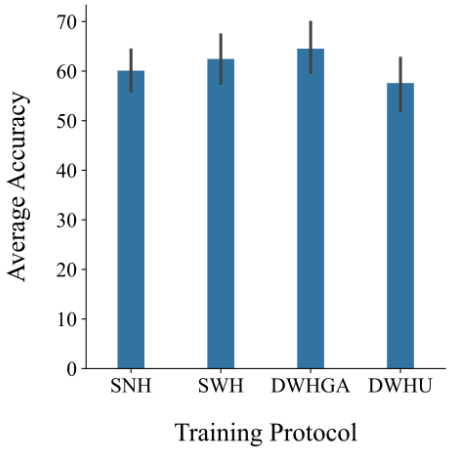


Fig. 10. Generalization ability of the classifiers. Each classifier is trained using the related protocol data and all data from the rest of the protocols are used as test data.

for intra-protocol validation to 53%. The maximum inter-protocol accuracy is for training using the DWHG and testing on the DWHU with 66%. Fig. 10 shows the average inter-protocol accuracy for all training protocols. It is visible from the plot that the DWHG and DWHU have the most and least generalization capabilities with 64% and 58%, respectively.

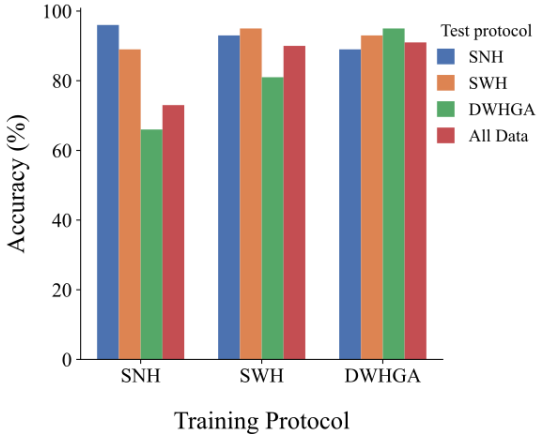


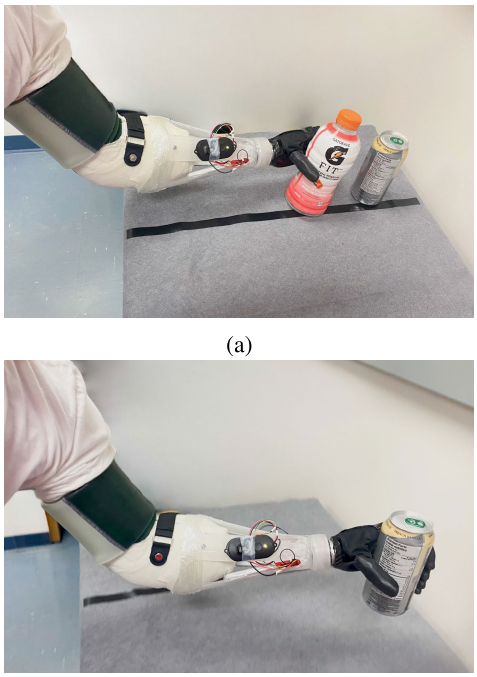
Fig. 11. Amputee subject performance validation. The last bar in each group shows the generalization ability of the related protocol.

D. Amputee Results

The offline analysis of the experiments with the amputee subject shows the same pattern as the healthy subjects. The intra-protocol validation shows the maximum accuracy for SNH with 96% followed by SWH (95%) and DWHG (94%). Regarding the generalizability of the models, DWHG has the maximum generalizability with 91% accuracy, followed by SWH (85%) and SNH (73%), which is also aligned with the results from non-disabled participants. The validation results for the amputee participant are shown in Fig. 11.

E. Real-Time Test

Five non-disabled participants and the amputee participant participated in the real-time test. Each of them was instructed to relocate seven objects, one after the other, from one side of one reference line to the other side of the other one, which is 55 cm apart, at their own pace. The models trained using 1. SNH, 2. SWH, and 3. DWHG data were used for this task. Any object falling between the lines during transfer was recorded as a failure, prompting the participant to proceed to the next object. Fig. 12 shows the amputee participant during the real-time test. Two key metrics, completion time and completion rate, were employed to evaluate the real-time performance of each control scheme. Completion time was measured from the moment the participant initiated the grasp of the first object until the last object was placed. The completion rate, on the other hand, represents the ratio of objects successfully transferred from one side to the other. The average measures across all participants are summarized in Table I, while Fig. 13 visually presents the performance of each training protocol for individual participants. As indicated in Table I, integrating the weight of the prosthetic hand into the training protocol resulted in a notable 24% improvement in the completion rate for non-disabled participants and 29% for the amputee participant. Additionally, further enhancements of 33% were achieved by incorporating dynamic arm movements, as compared to the baseline training protocol (without hand and dynamic movement), resulting in a successful transfer rate of 97.2% for the non-disabled participants. For the amputee participant, however, the dynamic arm movement did not improve the performance further. Additionally, the completion time



(a)

(b)

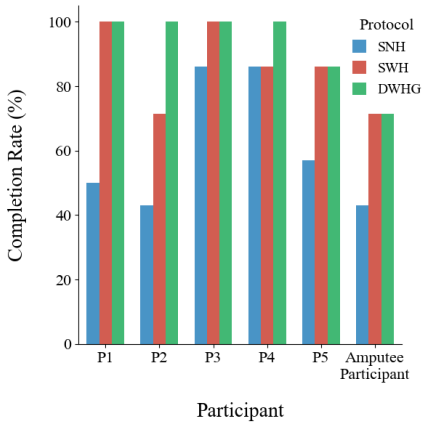
Fig. 12. Real-time test. Amputee subject **a)** grasping and **b)** transferring the objects.

decreased for both the SWH and DWHG protocols compared to the SNH protocol, with the DWHG protocol requiring slightly more time for both non-amputee and amputee participants. Demonstration videos of the real-time prosthetic control are available at: https://www.youtube.com/playlist?list=PLRhwx3gc_7yYxzDYuJ1GEyE-HR99a8dXa

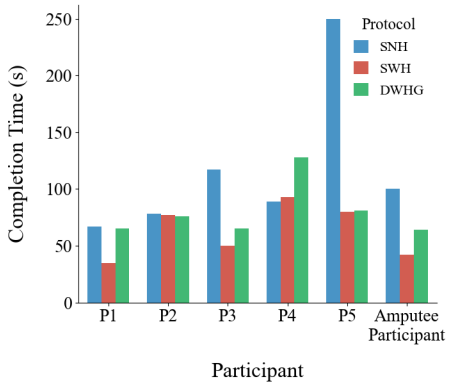
IV. Discussion

A PR-based EMG control system should be trained to suit real-world applications. Traditional methods, which rely on the models trained under highly controlled conditions, have limited applicability when it comes to real-life scenarios. Several studies have addressed real-world confounding factors, primarily muscle contraction variations and arm postures [8], [10], [13], [17], [31] by employing strategies such as advanced feature selection or machine learning models, and refined training datasets. These studies, however, often overlooked inadvertent EMG activation caused by external forces, like the force exerted by a rehabilitation device weight attached to a patient's hand that can disrupt EMG patterns. To support such a device, specific muscle contractions may be required, potentially interfering with EMG patterns crucial for motion decoding. To attain a robust classification, as demonstrated in this study, it's practical to collect training data while these devices are in use. Table II summarizes key related works, including those addressing external weight, that focus on tailoring training protocols to enhance system robustness.

The goal of this study was to investigate the impact of multiple confounding factors on myoelectric control and its practicality in real-world scenarios, rather than solely enhancing classifier accuracy. We aimed to compare the effects of more complicated data collection on model stability and generalizability.



(a)



(b)

Fig. 13. Real-time results for the 5 non-disabled and the amputee participants taking part in the real-time test **a)** real-time test completion rate for each participant. **b)** real-time test completion time for transferring all objects for each participant.

A. Weight Effect

Introducing the weight of the prosthetic hand during training led to a decrease in the intra-protocol offline accuracy, as it introduced more complexity to the signals; however, it helped in increasing the model generalizability and a significant improvement in the accuracy and robustness of the control scheme in real-time validation was observed. This suggests that incorporating real-world conditions, specifically the weight of the prosthetic hand, into training protocols can enhance the effectiveness of myoelectric PR control.

B. Arm Movement

We examined how dynamic arm movements affect the model performance by integrating arm position changes into the training protocols, in addition to incorporating the weight of the prosthetic hand. Although the intra-protocol offline accuracy remains lower than that of the baseline protocol (without prosthetic or arm movement), the model's generalizability increased even further compared to when only the weight was introduced. During the real-time test, this training protocol demonstrated either an equal or better completion rate for all participants (including the amputee participant); however, for almost all of them, it increased the

TABLE I
VALIDATION RESULTS FOR DIFFERENT TRAINING PROTOCOLS FOR INTACT LIMB (ITL) AND AMPUTEE (AMP) PARTICIPANTS

Training protocol		Offline Validation				Online Validation			
		intra-protocol		generalizability		completion time		completion rate	
		ITL	AMP	ITL	AMP	ITL	AMP	ITL	AMP
Static Training	SNH	**82.5%**	**96%**	60%	73.4%	120s	100s	64.4%	42.8%
	SWH	78.9%	95.5%	62.5%	85.2%	**67s**	**42s**	88.7%	**71.4%**
Dynamic Training	DWHG	79.9%	94.6%	**64.5%**	**90.7%**	83s	64s	**97.2%**	**71.4%**
	DWHU	79.6%	-	57.6%	-	-	-	-	-

TABLE II
STUDIES ADDRESSING DIFFERENT CONFOUNDING FACTORS IN MYOELECTRIC CONTROL

Study	Factors	Methodology	Participants	Testing Method	Physical Prosthetic Control	Accuracy/Error Rates and Key Findings
Scheme & Englehart [14]	Contraction strength	Ramp training scheme with dynamic contraction intensity	10 ITL	Offline and Real-Time tracking test	No	Inter-level error reduced from 31.2% to 18.9%
Gigli et al. [11]	Limb position and orientation	Static multi-position and dynamic data collection	14 ITL	Offline and Real-Time	Yes	Lower fatigue and shorter data acquisition time for dynamic acquisition
Teh & Hargrove [20]	Limb position, external load	Multi-position and dynamic classifiers with external loads	14 ITL, 5 AMP	Offline	No	Average accuracy increased from ~78% for the baseline to ~93% for multi-position classifier for ITL and from ~75% to ~88% for AMP
Teh & Hargrove [21]	Limb position, external load	Static and dynamic training data collection	14 ITL, 6 AMP	Real-Time using VR	No	Completion rate increased from 51.09% to 90.97% for ITL and from 61.98% to 83.33% for AMP
Yang et al. [19]	limb movement, contraction levels, inadvertent EMG activation	Four different training protocols where affecting factors are differently considered	8 ITL, 3 AMP	Offline and Real-Time	Yes	Intra-protocol RAR* decreased from 88.6% for the baseline to 82.2% for AMP, generalization RAR increased from 50.6% to 65.8%. Drop possibility reduced from 12.8% to 5.2% and completion time reduced by 8%
Our Study	Dynamic arm movement, prosthetic weight, guided and unguided training, automatic and manual labeling	Four training protocols addressing the effects of each factor	14 ITL, 1 AMP	Offline and Real-Time	Yes	Intra-protocol accuracy decreased from 96% for the baseline to 94.6% and generalization incresed from 73.4% to 90.7% for AMP. Completion rate increased from 64.4% to 97.2% for ITL and from 42.8% to 71.4% for AMP, Completion time decrease by 47% for ITL and 58% for AMP

ITL = Intact-Limb Participants, AMP = Amputee Participants
*ROC area ratio

completion time due to more unwanted gestures of the hand during the test. This increase in unintended gestures suggests that while the model can generalize better, it may also become more sensitive to minor variations in muscle signals, leading to less precise control.

C. Auto Labeling Vs. Video Labeling

We explored the reliability of auto-labeling versus video labeling in training data annotation. Our results indicated no significant difference in classification performance between the two labeling methods. This suggests that auto-labeling can be a practical and efficient approach for annotating training data, potentially reducing the time and effort required for manual annotation.

D. Guided Vs. Unguided Training

Additionally, we investigate the impact of guided versus unguided training protocols. Guided training involves providing participants with visual cues or instructions to perform specific gestures, for a specific amount of time, whereas unguided training allows participants to perform gestures without external guidance and at their own pace. Our study compares the effectiveness of these training protocols in terms of accuracy and generalization ability. As our findings suggest, having guidance during the data collection is beneficial, as it leads to better model generalizability.

V. LIMITATIONS AND FUTURE WORK

In this study, we examined the impact of various factors on the PR control of prosthetic hands. Our investigations

were first carried out with non-disabled volunteers, and then, to ensure the applicability of our findings, an amputee participant was also recruited to participate in the experiments. Although the general findings were consistent between the two groups, that would still be beneficial to recruit more amputee participants to make sure of the generalizability of the findings among the amputee group.

During our real-time tests, some participants reported discomfort when using the designed handle to perform a top grasp with the prosthetic hand and transfer objects. This was due to the uneven distribution of weight over the wrist and arm. As a next step, we intend to develop a more ergonomic handle for healthy volunteers, aiming to mitigate any factors that could potentially disrupt the results.

Additionally, another potential future direction is to explore the integration of an Inertial Measurement Unit (IMU) into the system. This integration would enable tracking of hand position, potentially leading to a more robust and precise control system.

VI. Conclusion

In this study, we evaluated the effect of prosthetic hand weight, arm movement, training methods, and different labeling strategies on the real-time PR control of the upper-limb prosthetic. We speculated that introducing more complexity to the training data will decrease the intra-protocol accuracy but increase the inter-protocol (the generalization of the model). Our experiments revealed that introducing prosthetic hand weight and arm movement to the training protocols can enhance the generalizability of the models to a great extent during the offline analysis, the intra-protocol accuracy, however, decreased as expected.

In contrast, employing video labeling does not offer an advantage over auto labeling; rather, it serves to validate its reliability. Our results also show that guided training yields a more robust classifier compared to the unguided protocol. To validate our offline findings, we selected the best models from the offline analysis and compared their real-time performance with the baseline protocol. Our real-time findings further support the offline results, demonstrating a more robust and reliable performance for training protocol involving the prosthetic hand. Introducing the arm movement also demonstrates a marginal improvement in the completion rate compared to the models only trained with the prosthetic hand, and a marked improvement compared to the baseline. This model, however, increases the completion time due to less consistent control compared to the previous protocol. Considering this trade-off between increased completion time and increased completion rate, and taking the complexity of the training data into account, the second protocol may be more advantageous for most use cases.

In comparison with current literature and to the best of our knowledge, there is no study that investigates the effect of the same confounding factors on the performance of prosthetic hands in a shared-control scheme, where EMG is used to trigger the automatic control part. However, there are similar studies which investigate the effect of weight, arm position, and movement in myoelectric control [18], [19], [20], [21]. Consistent with the findings of Cipriani et al. [18], our results confirmed that external weight and arm movements significantly degrade the performance of traditional classifiers. Additionally, we corroborated the observations of previous studies [11], [19], [20], [21], demonstrating that increasing EMG diversity during training sessions enhances the usability and generalization of PR-based EMG control schemes in real-world scenarios. However, as also observed in [19], [32], we found that while training under more realistic conditions enhances real-time performance, it can reduce intra-protocol offline accuracy, resulting in lower accuracies compared to the literature. As also highlighted in [19], this likely represents the practical performance of PR-based EMG control systems in clinical applications. These findings also underscore the limitations of offline metrics in fully capturing a system's practical effectiveness, emphasizing the need for performance validation in real-world scenarios.

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